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Modular Bio-Mechanical Actuator & Data Acquisition System

Objective: To design and build a modular motion-control system capable of (1) performing adhesive peel tests to predict skin pain and (2) acting as an automated XY/Z stage for digital microscopy.

Introduction to peel measurement:
https://www.youtube.com/watch?v=aQJ51U0D3wQ

1. System Architecture (The Modular Concept)

The project is split into three distinct modules:

1. **Motion control:** A Raspberry Pi 5 (4GB) + Stepper Driver + ADC in a single enclosure with "Plug & Play" connectors.
   NEMA 17 (High Torque)
   Raspberry Pi 5 (4GB)
2. **Peeling Module:** A vertical/horizontal rail with a load cell for adhesive testing.
3. **Imaging Module**: A Raspberry Pi Global Shutter camera with macro optics to visualize the "Peel Front" or biological samples.

2. Student Task List

Task 1: Electronics & "Brain" Integration

- Assembly: Wire the Raspberry Pi 4 to the TMC2208 Stepper Driver and the ADS1256 24-bit ADC.
- Firmware: Write a Python script to control motor speed in microns/s to mm/s.
- Modular Interface: Install GX16 Aviation Connectors on the control box so the motor and load cell can be swapped out easily.

Task 2: Mechanical Design (Peeling vs. Stage)

- Peeling Rig: Design a frame using 2020 Aluminum Extrusions. Mount a lead screw to move a carriage. The carriage must have a mounting point for a 10g Load Cell.
- Microscope Conversion: Design a 3D-printed adapter that allows the same NEMA 17 motor to attach to a Holmarc Manual Stage. This converts "Manual" to "Automated" for scanning samples.

Task 3: The "Pain Predictor" Vision System

- Optics: Mount the Raspberry Pi GS Camera. Students must determine the "Focal Distance" required to see adhesive interface
- Software: Use OpenCV to capture video at 30–120 fps.
- Synchronization: Create a script that overlays the Force (Newtons) onto the Video Frame in real-time.

Task 4: Data Analysis & Validation

- Experiment: Peel a standard medical tape from a silicone "skin" simulant.
- Analysis: Identify "Force Spikes" in the data and correlate them with "Skin Tenting" in the video.
- Metric: Calculate the Work of Adhesion (W = integral (Fds)).

Suggested website to find required items
https://robu.in/
https://www.holmarc.com/index.php

3. Bill of Materials

List items to be purchased

4. Expected Deliverables

1. A Functional Control Box: Capable of driving any NEMA 17 motor via Python.
2. Peel Test Report: A graph showing Force vs. Time synced with a video of the peel.
3. Modular Demonstration: Show the motor being moved from the Peel Rig to a Microscope Stage and functioning correctly.